Engineering Design and Feasibility of a Floating Photovoltaic (FPV) System in the UK

By Isaac Summers

Student number : 492991

1. Introduction

As global demand for renewable energy rises, solar photovoltaics (PV) remain among the fastest growing contributors to electricity generation, however large ground mounted systems require significant land, competing with agricultural, ecological and urban uses. Floating photovoltaic (FPV) systems overcome these limits by utilising underused water surfaces such as reservoirs and lakes. Studies have shown that FPV systems offer higher efficiency and reduced land impact, making it a promising emerging technology for renewable systems[1], [2].

FPV panels operate at lower temperatures through evaporative cooling, typically showing 5-10% higher output, they also limit evaporation and algae growth via partial shading which in turn improves water quality[3], [4]. Deploying FPV on just 1% of global manmade reservoirs could generate around 400GW[5], however, FPV can alter water-air interactions, shading, and thermal layers, potentially affecting freshwater ecosystems[3], mechanical wear, corrosion and biofouling remain long-term challenges for FPV[6]. Stable mooring, corrosion resistant materials, electrical safety and environmental monitoring are essential for their viability.

This report evaluates the engineering feasibility of a 50 kWp FPV array on a UK fresh water reservoir, applying multidisciplinary engineering methods to assess structural integrity, generation potential and economic viability. The selected site is Bewl water which is southeast England's largest inland reservoir, covering 3.1km^2 (309 ha) with an average depth of 9.9m aligning with recommended FPV deployment depths and is regulated by a siphon system[7], [8], [9]. Bewl water is located on the Kent-East Sussex border and therefore benefits from high solar irradiance due to Kent consistently being at the upper end of 750-1,110 kWh/m^2 annually, supporting year round FPV potential[10]. Its location is also within the UK power networks distribution region facilitating grid connection opportunities[11].

2. Methodology

2.1 Design Approach

The design of the Bewl Water 50 kWp floating photovoltaic array follows current international and national best practice standards for inland floating solar installations. Design references include DNV-RP-0584 for mechanical design[12] and BS IEC 62548 for electrical array design[13].

All calculations are performed analytically as recommended by these standards, CAD tools (CATIA v5) are used for the production of schematic and general arrangement drawings using BS 8888 for view layout, dimensioning and notation[14].

2.2 Conceptual System Design

2.2.1 Array Capacity and Layout

The FPV system uses 500Wp monocrystalline PERC PV modules (JA Solar JAM66S30/MR[15]), each 2.09m x 1.13m with weight of ~26.3 kg. Mono PERC was chosen for its improved

temperature coefficients, enhancing FPV performance where evaporative cooling lowers operating temperatures[16]. Achieving 50 kWp requires 100 500Wp modules in a 10 by 10 grid.

Following DNV-RP-0584 guidelines, the total array footprint includes the active area, walkways and edge clearances for safe access and maintenance, main walkways are 1m wide, with minimal column spacing and 0.5m perimeter clearance. The resulting footprint is:

$$Footprint = \big(10 * (2.09m + (0.5 * 2))\big) * (10 * (1 + 1.13m)) = 467m^2$$

Industry best practice limits FPV coverage to under 10% of reservoir area to protect water quality, this design covers only 0.015% of Bewl water's surface ($467m^2/3{,}090{,}000m^2$ * 100%)[17]. Other documented 50 kWp FPV systems can be seen to have a spatial footprint of ~500m^2 (0.13 acres) confirming the initial estimate[18].

2.2.2 Orientation and tilt

DNV-RP-0584 indicates that the incidence angle modifier (IAM) losses are primarily due to tilt and azimuth orientation, for Bewl water the most effective orientation is south because the sun's path is southerly in the northern hemisphere. The optimal tilt angle for fixed south facing PV panels in the United Kingdom is ~34deg reported by Jacobson and Jadhav (2018)[19]. However, for FPV systems as recommended by DNV a lower tilt angle should be considered to reduce wind and wave induced mechanical loads. Even if there is a small reduction in annual output, a lower tilt can actually increase summer energy capture due to the suns higher elevation. Consequently a tilt angle of 10-20 degrees is typical for FPV systems and a tilt angle of 15 degrees will be used for this project[20].

2.2.3 Floating Platform

The floating platform is a modular pontoon raft designed to support the array of 100 x 500Wp modules. For modularity, installation simplicity and maintenance, pontoons are arranged as continuous walkway rows forming a lattice structure, metal cross beams span between adjacent rows to support the PV modules, allowing each module load to be shared by two walkways with clear maintenance access. The pontoon unit can be described as follows:

- Geometry: 2.1m x 1m x 0.4m (L x W x H), matching the module dimensions.
- Material: HDPE provides high impact and tensile strength, excellent UV and thermal resistance, and long ESCR life which is consistent with DNV-RP-0584 guidance for a 25-30 year design life in freshwater[12], [21], [22].
- Interconnection: Pontoons are connected by steel plates and cross members to form rigid walkway rows, the cross members are also used to carry the PV modules.

2.3 Electrical connections and grid integration

The system integrates 100 PV modules into a string and inverter configuration whilst ensuring safety, reliability and ease of maintenance in a freshwater environment. Each module is certified to IEC 61215 and IEC 61730 for design qualification and safety[15], modules would be

mounted on the pontoons with IP67 rated junction boxes which are fully encapsulated to resist water ingress, humidity and mechanical movement induced by waves and wind[12].

Modules are arranged into 10 strings of 10 modules in series, producing a string voltage of ~ 384V DC at 13.04A under standard test conditions (calculated from max power voltage and current[15]) which would be within the acceptable voltage range of the selected inverter and combiner. Copper conductors with dual insulation connect each string module to IP65 rated DC combiner boxes in parallel.

A string inverter system has been chosen due to the moderate array size and proximity to the shore allowing installation on land close to the reservoir. The inverter converts DC from the strings to AC at 230V/50hz and feeds the output into the local distribution network through an onshore transformer (rated IP65+).

The system connects to the local UK distribution network through an onshore point of connection such as the water treatment facility in Bewl water[23]. A SCADA system following IEC 61724 should also be used to monitor array performance, inverter status and environmental conditions to provide supervision and fault detection where protective relays or disconnect switches can activate.

Lightning protection is implemented using air terminals and grounding rods, with fire detection and suppression systems included in the inverter and transformer enclosures in line with risk assessment guidelines. The structure is also bonded to the inverter earth and exposed conductive parts are terminated to earth rods at sufficient water depth minimising potential differences with the water body[12].

A basic electrical schematic can be seen below:

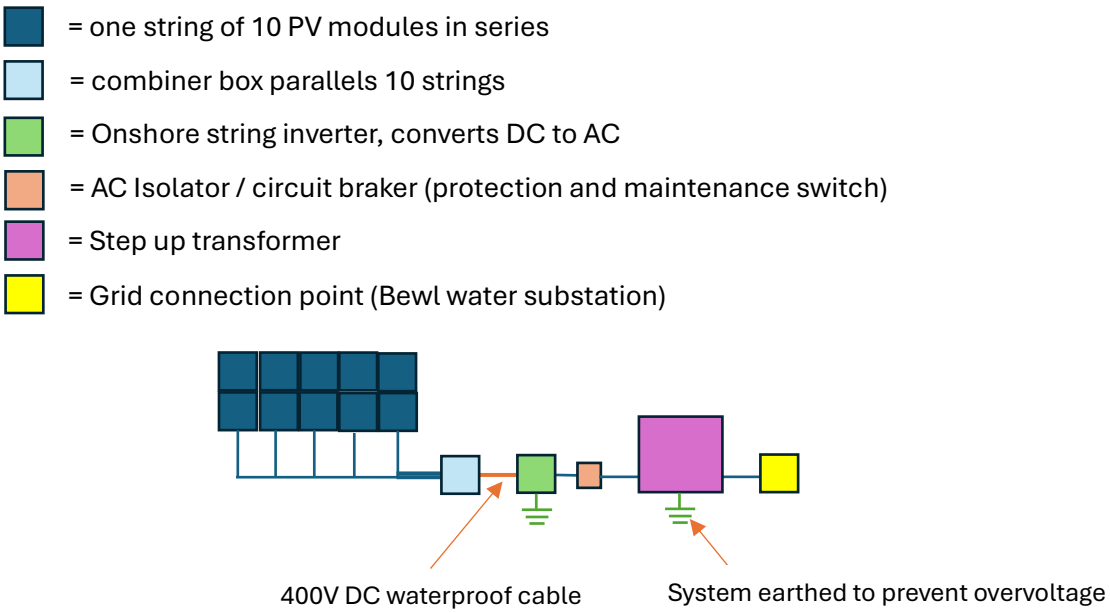


Figure 1 - Basic electrical schematic of the FPV system

3. Results and Discussion

3.1 Environmental and chemical analysis

Biofouling is a process in which micro and macro organisms accumulate on surfaces leading to the formation of biofilms which significantly accelerate material degradation due to trapped corrosive agents. This shows the need for antifouling strategies, one such strategy could be liquid infused coatings which have shown promise in reducing general biofouling in freshwater systems[24]. The designed FPV system uses HDPE which has high corrosion resistance and while considered more chemically stable than other polymers, can still show chemical leaching under harsh UV conditions which can pose potential health risks and toxicity, therefore a coating to protect against UV could be considered[25], [26].

FPV shading can improve water quality through reducing harmful algal blooms, which deplete oxygen levels, produce harmful toxins and disrupt the ecosystem. However, extensive FPV coverage can suppress beneficial photosynthesis and alter nutrient cycles[3], as discussed previously the designed system has only 0.015% coverage of Bewl water's surface area mitigating these risks.

FPV systems can influence aquatic ecosystems by providing floating structures that attract fish and other mobile species, potentially encouraging aggregative behaviour, fish may gather around FPV arrays due to shelter, feeding opportunities and the chance to encounter other fish. FPV structures could potentially increase local abundance and reproductive output however larger species such as birds may use the array for foraging or resting, whereas some sensitive species may avoid the FPV system potentially disrupting natural marine behaviour[3].

FPV's cool surface waters and reduce temperature variability below the panels which can alter vertical mixing and stratification in lakes and reservoirs, these changes could impact the distribution of oxygen and nutrients[3].

3.2 Mechanical and structural analysis

3.2.1 Buoyancy Calculations

Buoyancy can be verified using Archimedes' principle:

$$F_b = \rho g V_{submerged}$$

Where $\rho = 1000\ kg/m^3$, $g = 9.81\ m/s^2$, and $V_{submerged}$ is the submerged volume.

$$V_{sub} = L * W * d = 2.1 * 1 * 0.25 = 0.525\ m^3$$

$$F_{b,operational} = 1000 * 9.81 * 0.63 \approx 5.15\ kN$$

Each module plus mounting hardware weighs ~0.36 kN (26.3kg for the PV module + 10kg in rails, bolts and wiring), accounting for maintenance a load of 2kN/m^2 is assumed to represent a few people with tools. This total load is still below the ~5kN buoyant force of the pontoon therefore making sure the rafts float well above the water in normal conditions.

3.2.2 Stability and Metacentric Height

Stability is assessed using metacentric height (GM) per DNV recommendations for three interconnected pontoons:

$$GM = KB + BM - KG$$

Where:

- KB = centre of buoyancy above bottom of the pontoon ≈ 0.125 m (d/2)
- BM = I/V = metacentric radius based on the second moment of area (I) and submerged volume (V), for a rectangle $I = \frac{B*L^3}{12} \rightarrow \frac{3*(2.1)^3}{12} \approx 2.44m^4, V = 3 * 2.1 * 0.25 = 1.575m^3 \rightarrow BM = \frac{I}{V} = \frac{2.44}{11.575} \approx 1.55m$
- KG ≈ 0.6m above the pontoon base (conservative estimate using combined mass and height of HDPE float, frame and tilted PV module)

$$GM = 0.125 + 1.55 - 0.6 \approx 1.075m$$

A positive GM indicates strong static stability under normal wind and wave conditions[27], and the full raft with multiple connected rows will show even greater stability due to its larger area.

3.2.3 Wind and Wave loads

Wind and wave calculations determine the horizontal forces on the FPV array and resulting mooring line tension to ensure correct configuration for ultimate and accidental limit states.

The projected wind area per row is:

$$h_{eff} = pontoon\ height + module\ height(with\ tilt) = 0.4 + 1.13\sin(15^\circ) = 0.692\ m$$

$$A_{module} = h_{eff} * module\ width = 0.692 * 2.09 = 1.447\ m^2$$

$$A_{row} = 10A_{module} = 14.47m^2$$

The wind force can be given by:

$$F_{wind,row} = 0.5\rho_{air}V^2C_dA_{row} = 6.65\ kN$$

Where V = 25m/s representing a conservative estimate based on a 30 year return speed of 23m/s across the UK[28]. The module row is modelled as a flat plate with a maximum to characteristic area ratio of 2, as the module may tilt slightly increasing the effective area exposed to the wind therefore a drag coefficient (C_d) of ~1.2 is used [29].

Wave force can be estimated using the Morison equation[30], the significant wave height H ≈ 0.2m, was predicted using the SMB model[31], which uses the sites fetch (3.42 km[7]) and average 10m wind speed (5.29 m/s in a 9km^2 area around Bewl water[32]), a conservative H=0.5m is used to include extreme events, wave period (T) is estimated as ~1.8s[31] and site water depth (d) as 10m. Pontoon height (D) is 0.4m and row width 2.1m x 10 modules. Based on low Keulegan Carpenter (KC) numbers (which is a dimensionless number representing flow regime), the drag (C_d) and inertia (C_m) coefficients are estimated as 2 and 1.8 respectively[33]. These coefficients are preliminary and should be validated by simulation or physical testing.

The wavelength (λ) and wave number (k):

$\lambda = \frac{gT^2}{2\pi}\tanh\left(\frac{2\pi d}{\lambda}\right) = 5.06\text{m}, \quad k = \frac{2\pi}{5.06} = 1.24\,\frac{rad}{m}$, confirming deep waves (d≥0.5$\boldsymbol{\lambda}$):

For deep waves the horizontal velocity (u) and acceleration ($\dot{u}$) can be described as:

$$u = \frac{\pi H}{T}e^{kd}\sin(-), \quad \dot{u} = \frac{2\pi^2 H}{T^2}e^{kd}\sin(-)$$

Substitution into the equations for drag and inertia force:

$$F_D = 0.5\rho C_d D\left(\frac{\pi H}{T}\right)^2 \frac{1}{2k}\left(1 - e^{-2kd}\right)\cos(kx - \sigma t)|\cos(kx - \sigma t)|$$

$$F_I = \rho C_m \frac{\pi D^2}{4}\frac{2\pi^2 H}{T^2}\frac{1}{k}\left(1 - e^{-kd}\right)sin(kx - \sigma t)$$

$$F_D = 122.63\cos(kx - \sigma t)|\cos(kx - \sigma t)|, \quad F_I = 554.77sin(kx - \sigma t)$$

Total wave force:

$$F_{T,wave} = 122.63\cos(kx - \sigma t)|\cos(kx - \sigma t)| + 554.77sin(kx - \sigma t)\ N$$

The inertial force dominates and maximum occurs at the peak of the sine term, therefore, for a 1m pontoon section $F_{max,wave} \approx 554$N and 11.65 kN over the full 21m raft width in extreme conditions.

3.3 Anchoring and mooring system design

The anchoring and mooring system maintains the FPV rafts position and limits horizontal drift under operational and extreme conditions, the design follows DNV-RP-0584 mooring design guidance[12].

To account for uncertainty at the ultimate limit state (ULS) the design tension (T_d) is:

$$T_d = y_{mean} * T_{c,mean} + y_{dyn} * T_{c,dyn}$$

With mean wind force of 0.298 kN and pretension of 1 kN, $T_{c,mean}$ = 1.298 kN. Dynamic tension from gust wind and wave loads give $T_{c,dyn}$= 11.65+(6.65-0.298)=18kN. Load factors for ULS are y_{mean}= 1.3 and y_{dyn}=1.75, resulting in T_d= 33.2kN.

Following DNV recommendations the system is a multi-line symmetric catenary with vertical line angle under 25° giving line tension of T_d/cos(25) = 36.6 kN. HMPE rope (6mm) is selected for high strength, UV resistance and low elongation, with breaking load of 33.3 kN[34], using a safety factor of 2 the working load per line is 16.65 kN.

Therefore, 36.6/16.65 = ~2.2, so at least 3 lines are needed. 8 lines are used for symmetry and uniform loading if a line fails, each carries 4.58 kN which is well below the working load ensuring safety under extreme conditions.

The system will be secured using submerged concrete gravity anchors due to their simplicity, inexpensiveness, sites shallow water depth and proven holding capacity. Anchor weight is

designed to resist the maximum design line tension (T_d) including a material factor of $y_m = 1.3$ for ULS[12]. The holding capacity of a submerged gravity anchor is approximately equal to its submerged weight:

$$R_d = W_{sub} = W_{anchor} - \rho_{water} * V_{anchor} * g$$

Where W_{anchor} is $\rho_{concrete} * V_{anchor} * g$, $\rho_{concrete} = 2400\ kg/m^3$

$$R_d = W_{sub} \geq T_{line} * y_m = 4.58 * 1.3 \approx 5.95\ kN$$

$$V_{anchor} = \frac{W_{sub}}{(\rho_{concrete} - \rho_{water}) * g} \approx 0.44 m^3$$

Therefore, each anchors should be 0.44m^3 of concrete (~1 ton) and attached to each individual mooring line.

3.4 Technical schematics

Technical drawings of the system are provided below, showing the full FPV system including the mooring and anchoring system and a smaller sub section highlighting the major components:

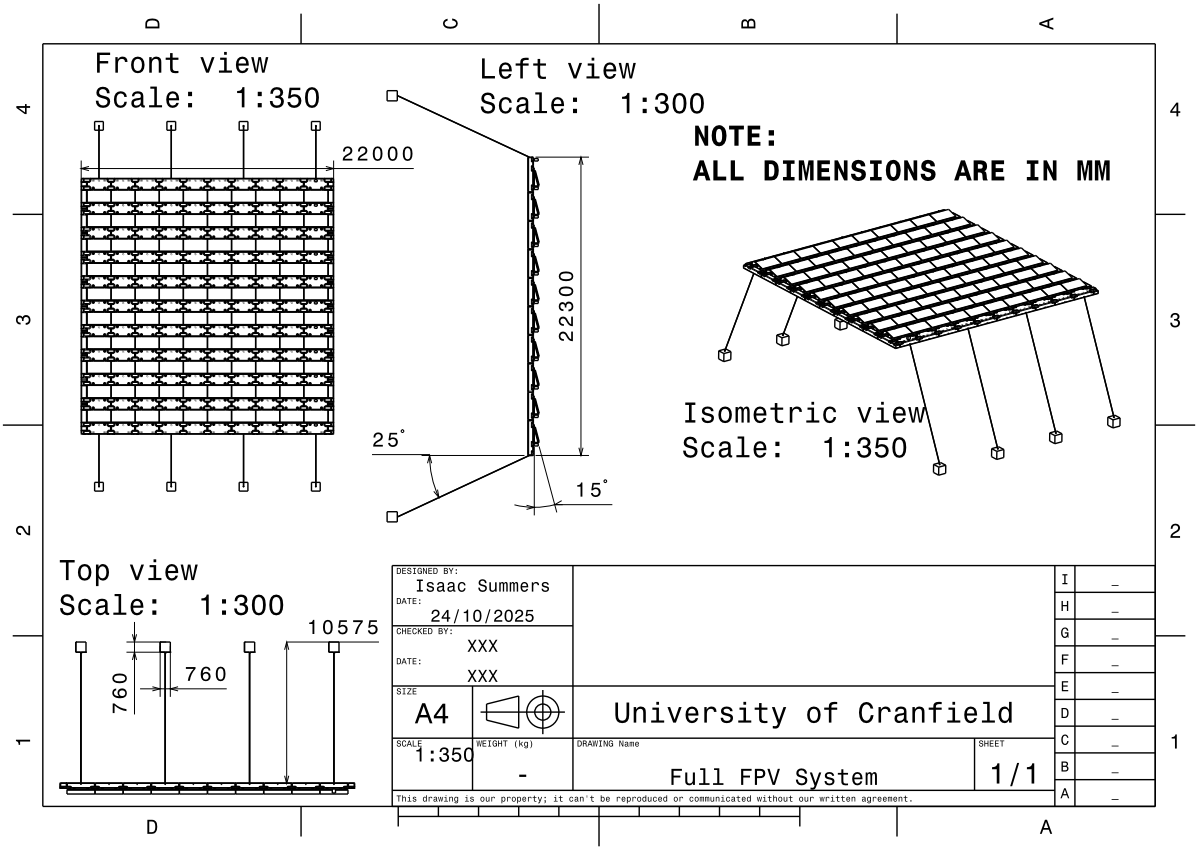


Figure 2 - Full system technical drawing

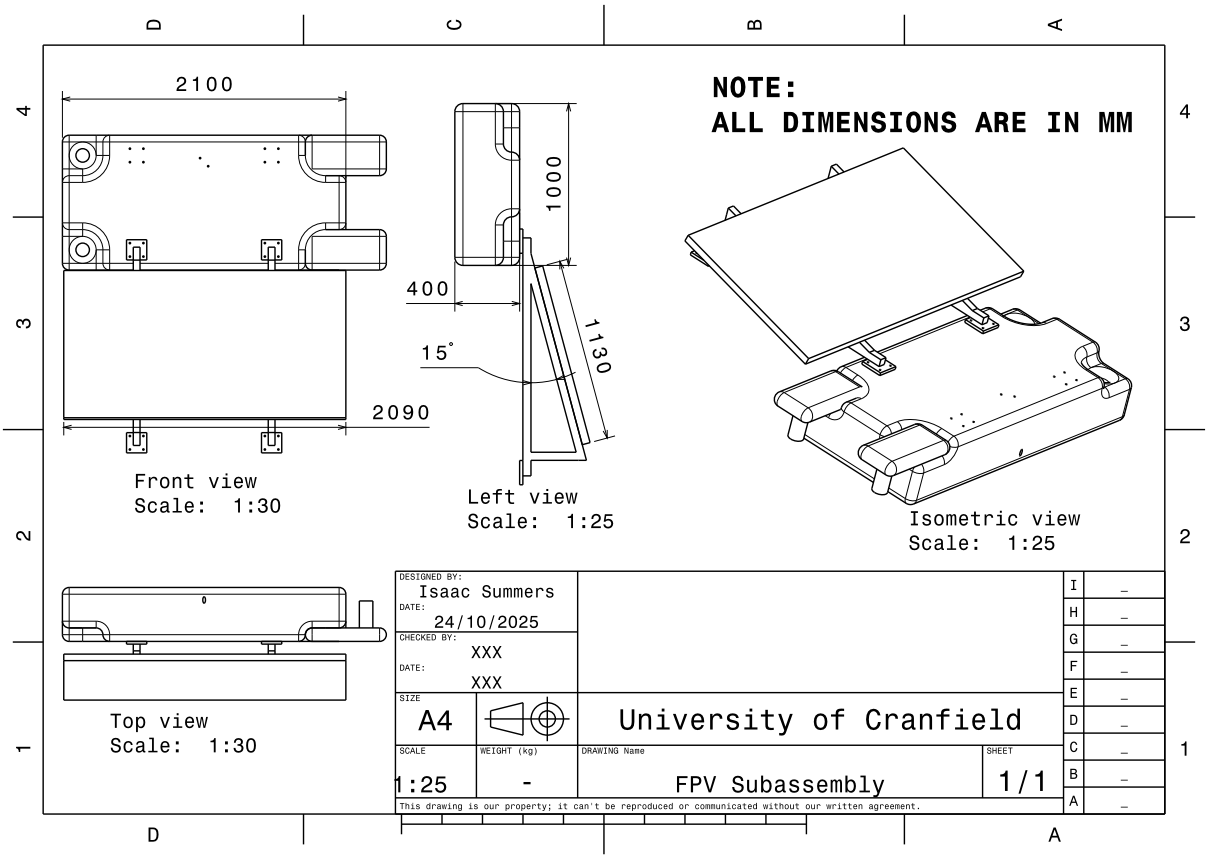


Figure 3 - system sub section technical drawing

3.5 Performance and carbon analysis

3.5.1 Cost estimate, performance and payback period

To estimate system performance a conservative average irradiance of 950 kWh/m^2 in Kent and system performance ratio of 0.85 is used (average expected performance including losses for FPV systems from figure 55 in[35]), this includes different forms of losses commonly associated with FPV systems such as optical losses, module and thermal losses, inverter and electrical losses, mismatch losses, system degradation, etc[12]. The annual energy output can be estimated based on the following equation (IEC 61724-1):

$$PR = \frac{E_{out}/P_0}{H_i/G_{ref}} \rightarrow E_{out} = 50kWp * \frac{950kWhm^{-2}}{1000Wm^{-2}} * 0.85 = 40.375\ MWh/year$$

Where PR is the performance ratio, E_{out} is the annual energy output (kWh), P_0 is the installed DC power (kWp), H_i = in plane annual irradiance (kwh/m^2), G_{ref} is the reference irradiance (1000 W/m^2).

The system is estimated to cost £1724/kWp accounting for a 25% increase over ground mounted systems[36], [37], giving a total cost of £86,200 for the 50kW system. With annual generation of 40,375kWh and electricity cost of £0.2573/kWh[38], annual revenue is £10,388. After assuming annual operation and maintenance costs equal to 1% of the total cost[39], the net annual profit is £9,527 resulting in a payback of 9 years.

3.5.2 Carbon analysis

Using a UK grid carbon intensity of 125gCO_2/kWh[40],the annual carbon savings are $40{,}375kWh * 0.125kgCO_2/year \approx 5.05\ tCO_2/year$, over a 25 year lifespan this amounts to approximately 126tons of CO_2 avoided, demonstrating the systems contribution to decarbonising electricity supply.

4. Conclusions and recommendations

The Bewl water 50 kWp floating solar PV system demonstrates the feasibility and efficiency gains through multidisciplinary integration of mechanical, electrical, and environmental engineering disciplines. Modular pontoons deliver stability with a positive GM, while electrical and environmental decisions such as reduced panel tilt and limited water coverage strike a balance between maximizing output, ecosystem protection and maintenance needs. The system achieves measurable carbon savings of approximately 5tCO_2 annually, directly supporting the UK's net zero targets and sustainable energy transition and a system payback period of only 9 years.

For ongoing improvement further integration of SCADA driven monitoring and antifouling technology is recommended to maximise performance, reliability and water quality, future studies should also refine wind and wave assessments and further investigate advanced materials for corrosion control. With continuous innovation in design and operational strategies large scale FPV arrays can substantially contribute to ambitions for a resilient net zero electricity grid.

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